

Lecture 17

Trefoil v2: Booleans, If Expressions, Bindings

Group Work

```
(* (if p t e) should evaluate to `e` if `p`  
evaluates to `false`, and to `t` otherwise *)  
| If (pred, thn, els) -> (  
    match interpret env pred with  
    | Bool false -> interpret env els  
    | _ -> interpret env thn)
```

For each of the following alternate semantics of `if` expressions, write out how the implementation would change.

1. `(if p t e)` should evaluate to the result of evaluating `e` if `p` evaluates to `false` or `0`, and to the result of evaluating `t` if `p` evaluates to any other value.
2. `(if p t e)` should evaluate to the result of evaluating `e` if `p` evaluates to `false`, to the result of evaluating `t` if `p` evaluates to `true`, and throw an error if `p` evaluates to an int.

Write 2-3 Trefoil programs and their expected result under each version of the semantics.

Group Work

Assume `parse` is a function that correctly parses a string to a binding

```
let e0 = StrMap.empty
let e1 = interpret_binding e0 (parse "(define x (+ 1 2))")
let e2 = interpret_binding e1 (parse "(define y (* x x))")
let e3 = interpret_binding e2 (parse "(+ x y)")
let e4 = interpret_binding e3 (parse "(define x (- y 1))")
let e5 = interpret_binding e4 (parse "(let ((x 100)) (+ x 10))")
let e6 = interpret_binding e5 (parse "(define y (let ((x 1)) (+ x y)))")
```

What are `e1` - `e6` bound to after this code is executed?